

SME Business Intelligence & Management Decision Support System

Technical Project Documentation

1. Project Purpose

This project demonstrates an end-to-end workflow for converting transactional data into information that can support management decisions. The objective is to demonstrate the progression from technical data skills to business decision support: cleaning and engineering data, analysing it, identifying business patterns, and translating those patterns into management recommendations.

2. Tools and Technologies

Python, Pandas, PostgreSQL, SQL, Power BI and GitHub.

3. Data Pipeline

Raw transaction data → Python/Pandas cleaning → PostgreSQL → SQL analysis → Power BI → management insights and recommendations.

4. Data Quality Validation

500 records were loaded. Required-field checks returned 500/500. Duplicate transaction IDs: 0. Invalid quantities: 0. Revenue/cost/profit calculation inconsistencies: 0.

5. Executive KPI Results

500 transactions; 10,904 units sold; ■695,582,508.00 revenue; ■561,713,050.00 total cost; ■133,869,458.00 total profit; 19.25% profit margin.

6. Key Product Insight

Granite 20 Tons generated ■465,302,500 revenue and ■89,342,500 profit, representing 66.89% of total revenue. PVC Pipe 1in recorded the highest product-level margin at 27.86%.

7. Category Insight

Aggregates generated ■633,751,000 revenue and ■120,271,000 profit at 18.98% margin. Plumbing recorded a 25.35% margin and Blocks 25.29%.

8. Customer Insight

Real Estate customers generated ■282,338,449 revenue and ■55,497,399 profit. Construction Companies generated ■240,545,069.50 revenue and ■45,600,369.50 profit.

9. Time-Based Insight

January recorded the highest monthly revenue at ■182,224,076.50 and profit of ■34,708,226.50. March recorded the lowest revenue at ■55,190,717 and lowest margin at 18.60%. June recorded the highest margin at 19.97%.

10. Management Recommendations

Monitor dependence on Granite 20 Tons; investigate high-margin categories; maintain strong relationships with major customer segments; investigate the March decline; and use revenue and margin together when making product decisions.

11. Power BI Stage

The next stage will convert the SQL results into an executive dashboard showing KPIs, monthly trends, product/category performance, customer performance, revenue concentration and management recommendations.

12. Project Significance

The project demonstrates that business intelligence is not only about producing charts. It connects data engineering and analysis with management interpretation and action.

Important note

This is a portfolio/demonstration project based on 500 transaction records. The figures demonstrate the analytical workflow and should not be represented as audited financial statements or as records belonging to a specific real business.